1. Create a login form with username and password fields, and validate non-empty input.

XML CODE:

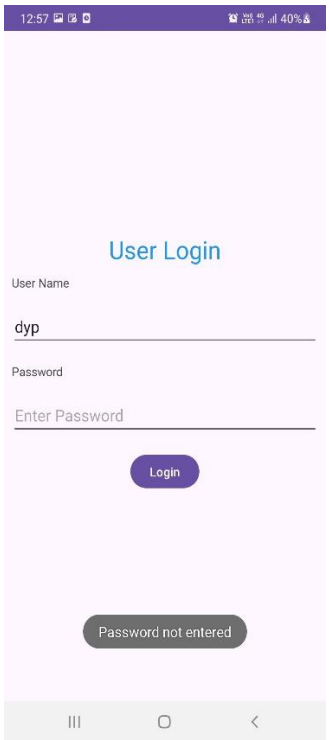
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:orientation="vertical"
   android:gravity="center"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
        android:text="User Login"
        android:textSize="30dp"
        android:textColor="#2196F3"
         />

    <TextView
        android:id="@+id/textView2"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
        android:layout_gravity="left"
        android:layout_margin="10dp"
        android:textSize="15dp"
        android:text="User Name" />

    <EditText
        android:id="@+id/editTextText"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
        android:ems="10"
        android:layout_margin="10dp"
        android:inputType="text"
        android:hint="Enter User Name"
        android:text="" />

    <TextView
        android:id="@+id/textView3"
android:layout_width="match_parent"
        android:layout_margin="10dp"
        android:textSize="15dp"
android:layout_height="wrap_content"
        android:text="Password" />

    <EditText
android:id="@+id/editTextTextPassword"
android:layout_width="match_parent"
android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:ems="10"
        android:hint="Enter Password"
       android:inputType="textPassword" />

 <Button
        android:id="@+id/btnLogin"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:text="Login" />
</LinearLayout>
```

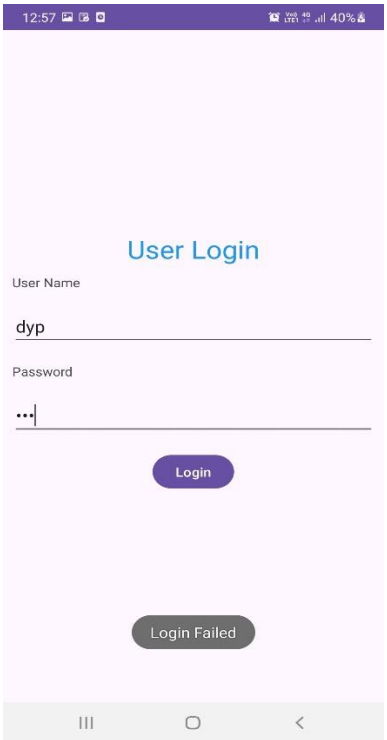
JAVA CODE:

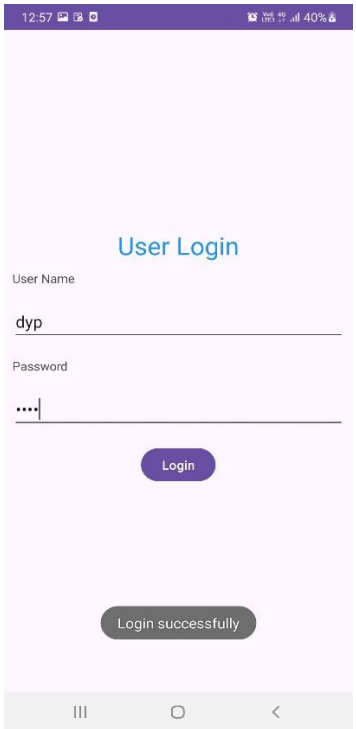
```
package com.example.app8loginui;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;

 import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;


import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        EditText edtUNm = findViewById(R.id.editTextText);
        EditText edtPw = findViewById(R.id.editTextTextPassword);
        Button btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String strUserNm = edtUNm.getText().toString().trim();
                String strPass = edtPw.getText().toString().trim();

                if(strUserNm.isEmpty()) {
                    Toast.makeText(getApplicationContext(), "User Name not entered",
Toast.LENGTH_LONG).show();
                    return;
                }else if (strPass.isEmpty())
                {
                    Toast.makeText(getApplicationContext(), "Password not entered",
```

```
Toast.LENGTH_LONG).show();
                return;
            }

            if (strUserNm.equals("dyp") && strPass.equals("comp")) {
                Toast.makeText(getApplicationContext(), "Login successfully",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(getApplicationContext(), "Login Failed",
Toast.LENGTH_SHORT).show();
            }

        }
    });
  }
}
```

OUTPUT:







2. Add gender selection using radio buttons and display the selected gender on login.
3.Implement a "Remember Me" checkbox and show a toast message based on its state.
4.Design a form that disables the login button until all fields are filled. 5.Modify the form to include a progress bar that appears during login processing.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

  <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="User Login"
        android:textSize="30dp"
        android:textColor="#2196F3"
         />

     <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="left"
        android:layout_margin="10dp"
        android:textSize="15dp"
        android:text="User Name" />

     <EditText
        android:id="@+id/editTextText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:layout_margin="10dp"
        android:inputType="text"
        android:hint="Enter User Name"
        android:text="" />

<RadioGroup
    android:id="@+id/rgb"
    android:gravity="left"
    android:layout_width="300dp"
    android:layout_height="100dp" >

    <RadioButton
       android:id="@+id/rdbtnMale"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Male" />

    <RadioButton
       android:id="@+id/rdbFemale"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Female" />
</RadioGroup>

<CheckBox
    android:id="@+id/chkBoxRem"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Remember Me" />
```

```
    <TextView
        android:id="@+id/textView3"
        android:layout_width="match_parent"
        android:layout_margin="10dp"
        android:textSize="15dp"
        android:layout_height="wrap_content"
        android:text="Password" />

    <EditText
        android:id="@+id/editTextTextPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:ems="10"
        android:hint="Enter Password"
        android:inputType="textPassword" />

    <TextView
        android:id="@+id/textView3"
        android:layout_width="match_parent"
        android:layout_margin="10dp"
        android:layout_height="wrap_content"
        android:text="Select Gender" />

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:enabled="false"
        android:text="Login" />

    <TextView
        android:id="@+id/txtGenMsg"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="" />

    <ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyle"
        android:visibility="invisible"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

</LinearLayout>
```

JAVA CODE:

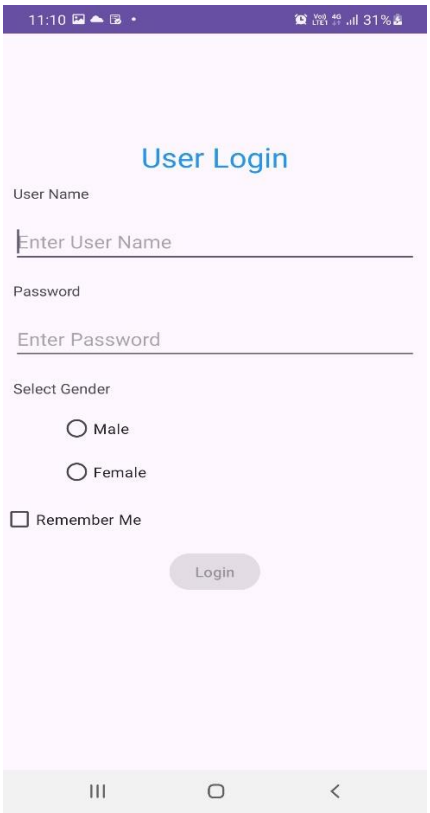
```
package com.example.app8loginui;

import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.text.Editable;
import android.text.TextWatcher;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.ProgressBar;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
```

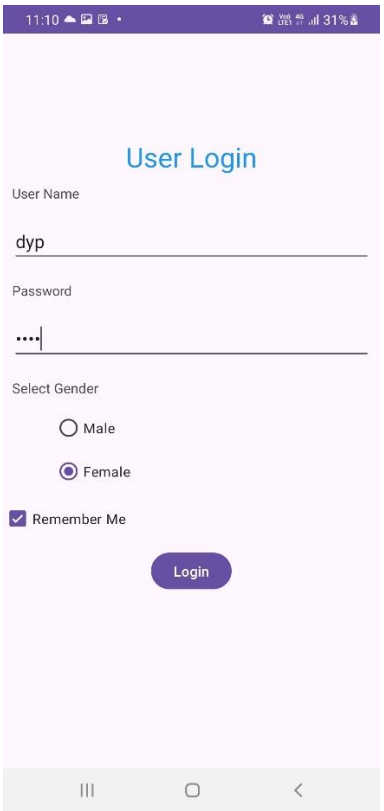
```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        EditText edtUNm = findViewById(R.id.editTextText);
        EditText edtPw = findViewById(R.id.editTextTextPassword);
        Button btnLogin = findViewById(R.id.btnLogin);
        RadioButton rbM = findViewById(R.id.rdbtnMale);
        RadioButton rbF = findViewById(R.id.rdbFemale);
        RadioGroup rg = findViewById(R.id.rgb);
        TextView txt = findViewById(R.id.txtGenMsg);
        CheckBox chk = findViewById(R.id.chkBoxRem);
        ProgressBar pr = findViewById(R.id.progressBar);

        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                pr.setVisibility(View.VISIBLE);

                Handler hd = new Handler(Looper.getMainLooper());
                hd.postDelayed(new Runnable() {
                    @Override
                    public void run() {
                        pr.setVisibility(View.INVISIBLE);

                        String strUserNm = edtUNm.getText().toString().trim();
                        String strPass = edtPw.getText().toString().trim();

                        if(strUserNm.isEmpty()) {
                            Toast.makeText(getApplicationContext(), "User Name not entered",
Toast.LENGTH_LONG).show();
                            return;
                        }else if (strPass.isEmpty())
                        {
```

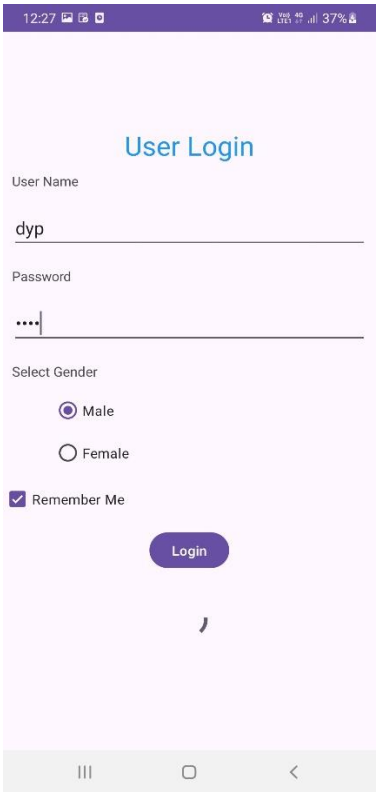
```
                Toast.makeText(getApplicationContext(), "Password not entered",
Toast.LENGTH_LONG).show();
                return;
            }

            if (! rbM.isChecked() && ! rbF.isChecked()) {
                Toast.makeText(getApplicationContext(), "Gender not selected",
Toast.LENGTH_LONG).show();
                return;
            }
            else
            {
                if (rbM.isChecked())
                    txt.setText("Selected Gender: Male");
                else
                    txt.setText("Selected Gender: Female");
            }

            if (chk.isChecked())
                Toast.makeText(getApplicationContext(), "Remember Me is selected",
Toast.LENGTH_SHORT).show();
            else
                Toast.makeText(getApplicationContext(), "Remember Me is not selected",
Toast.LENGTH_SHORT).show();


            if (strUserNm.equals("dyp") && strPass.equals("comp")) {
                Toast.makeText(getApplicationContext(), "Login successfully",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(getApplicationContext(), "Login Failed",
Toast.LENGTH_SHORT).show();
            }

        }
    }, 5000);

    }
});
```

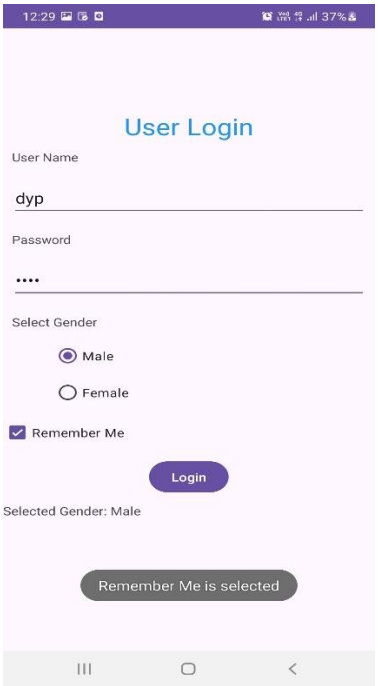
```
        edtUNm.addTextChangedListener(new TextWatcher() {
            @Override
            public void afterTextChanged(Editable s) {
                if ( ! edtUNm.getText().toString().isEmpty() &&
                        ! edtPw.getText().toString().isEmpty() )
                {
                    btnLogin.setEnabled(true);
                }
            }

            @Override
            public void beforeTextChanged(CharSequence s, int start, int count, int after) {
            }
            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
            }
        });

        edtPw.addTextChangedListener(new TextWatcher() {
            @Override
            public void afterTextChanged(Editable s) {
                if ( ! edtUNm.getText().toString().isEmpty() &&
                        ! edtPw.getText().toString().isEmpty() )
                {
                    btnLogin.setEnabled(true);
                }
            }

            @Override
            public void beforeTextChanged(CharSequence s, int start, int count, int after) {
            }
            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
            }
        });
    }
}
```

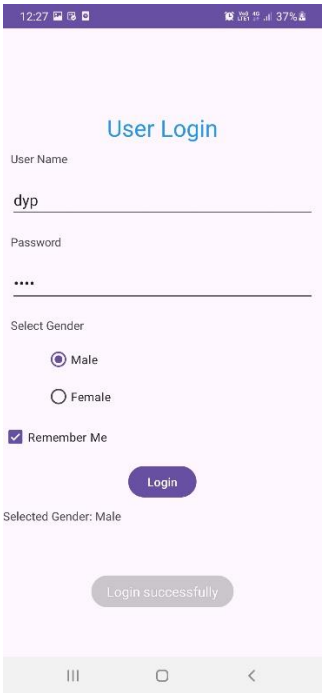
OUTPUT:


11:10
31%
User Login
User Name
Enter User Name
Password
Enter Password
Select Gender
Male
Female
Remember Me
Login


11:10
31%
User Login
User Name
dyp
Password
Select Gender
Male
Female
Remember Me
Login


12:27
37%
User Login
User Name
dyp
Password
Select Gender
Male
Female
Remember Me
Login


12:29
37%
User Login
User Name
dyp
Password
Select Gender
Male
Female
Remember Me
Login
Selected Gender: Male
Remember Me is selected


12:27
37%
User Login
User Name
dyp
Password
Select Gender
Male
Female
Remember Me
Login
Selected Gender: Male
Login successfully